Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

Course Introduction
Video • 1 min

Course Overview
Reading • 2 min

Syllabus
Reading • 10 min

Meet Your Instructor
Video • 1 min

Meet Your Fellow Learners
Discussion Prompt • 10 min

Academic Integrity
Reading • 1 min

Lesson 1: Introduction to Linear Algebra

Introduction to Machine Learning
Reading • 25 min

Introduction to Linear Algebra
Reading • 4 min

Why Linear Algebra and Mathematics?
Reading • 2 min

Notation
Reading • 5 min

Lesson 2: Systems of Linear Equations

Foundational Concepts of Systems of Linear Equations
Reading • 20 min

[H5P] System of Linear Equations
Ungraded App Item • 20 min

Solved Example and Recommended Resources
Reading • 20 min

Check Your Knowledge: System of Linear Equations
Graded Assignment • 16 min

Lesson 3: Matrices

Matrices and Matrix Operations
Reading • 20 min

Matrices
Video • 8 min

Helpful Resources and Solved Examples
Reading • 10 min

Check Your Knowledge: Matrices
Graded Assignment • 16 min

Lesson 4: Vector Spaces

Foundations of Vector Spaces: Operations and Subspaces
Reading • 20 min

Vector Space
Video • 7 min

Vector Space Properties
Reading • 10 min

Subspace
Reading • 1 min

Check Your Knowledge: Vector Spaces
Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

Introduction to Orthogonal Complement
Reading • 20 min

Introduction to Orthogonal Projections
Reading • 40 min

Importance of Projections
Reading • 4 min

Projection Onto One-Dimensional Subspaces (Lines)
Reading • 15 min

Projection Onto General Subspaces
Reading • 10 min

Projections Case Study
Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Syllabus

0:18 / 8:04 2x

Course Description

This course provides a comprehensive introduction to key probabilistic concepts and optimization techniques used in machine learning. You'll learn to apply the sum rule, product rule, and Bayes' theorem to solve complex problems and estimate hidden variables from observed data using Bayesian inference. The course also covers strategies to minimize error and loss in machine learning models through optimization techniques while exploring methods to enhance model accuracy and efficiency. Additionally, you'll assess model performance and generalizability using cross-validation approaches, equipping you with the skills to develop and evaluate robust machine learning models.

Course Outcomes

Upon successful completion of this course, you will be able to:

Course Overview

- Apply the sum rule, product rule, and Bayes' theorem to solve problems in machine learning.
- Estimate hidden variables from observed data using Bayesian inference.
- Minimize the error or loss of machine learning models through various optimization techniques.
- Assess model performance and generalizability using different cross-validation techniques.
- Improve model accuracy and efficiency by exploring and implementing various optimization methods.

Course Materials

The link to reading materials and resources to learn on the topics can be found in each week's learning module. All materials are available online for free, no required resources need to be purchased. There is no required textbook to supplement the course materials. *Note: Beware that some resources may open in a new tab.*

Course Outline

The course consists of 4 modules that focus on the following key areas:

Module	Learning Objectives
1 — Introduction to Linear Algebra and Vector Spaces	• Explore the role of mathematics in the science of machine learning • Identify the properties and requirements of vector spaces and subspaces, including vector addition, scalar multiplication, and the criteria for subspaces • Explore systems of linear equations, graphs, and matrix forms
2 — Linear Transformation and Matrix Decomposition	• Apply the concepts of linear mappings and their properties. • Interpret the concepts of eigenvectors and eigenvalues. • Explore Cholesky decomposition and its applicability to symmetric, positive, definite matrices. • Explore singular value decomposition.
3 — Probability Foundations for Statistical Learning	• Explore the sum rule, product rule, and Bayes' theorem, and their applications in machine learning. • Interpret Bayesian inference, including the posterior distribution and its applications in machine learning.
4 — Introduction to Model Evaluation and Optimization	• Minimize the error or loss of a model through optimization techniques in machine learning • Explore different cross-validation techniques to assess model performance and generalizability • Explore various optimization techniques to improve model accuracy and efficiency

Course Structure and Learning Activities

There are 4 modules in this course and each module may take about 4-5 hours to complete. Each module represents 1 week of learning with corresponding assignments per module. You can advance at your own pace; it is recommended that you take no longer than a week to complete each module. A consistent pace will help you complete the module and move on to the next course in the sequence.

This course is comprised of the following elements:

- **Videos:** In each module the concepts you need to know will be presented through a collection of short videos and/or an interactive presentation. You may stream these videos for playback within the browser by clicking on their titles.
- **Check Your Knowledge Quizzes:** Each module will include some quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each quiz. There is no time limit on how long you take to complete each attempt at the quiz.
- **Readings**: Each module may include several required and/or supplemental readings. The supplementary materials are provided for you to help you to further explore the course topics.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. Only those who complete and submit all assignments will receive a certificate of completion of this course.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera Financial Aid is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, please see the help topics here.

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	1-3 per module	.25-.5 hour	N/A	N/A
Interactive Presentations	Yes	1 per module	.5 hour	N/A	N/A
Check Your Knowledge Quizzes	Yes	4-7 per module	.5 hour	80%	100%

Getting and Giving Help

- Use the Learner Help Center to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with Coursera's policy on Accessibility.

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to Coursera's Honor Code, we also have high expectations for conduct during course participation.

Academic Code of Conduct

Above all else, learners are expected to ensure that their conduct helps to create an atmosphere conducive to learning and the interchange of knowledge. While it is understood that some of these items are subject to interpretation, learners should nonetheless endeavor to:

- Be respectful of fellow learners.
- Not discriminate against fellow learners in any manner.
- Conduct peer reviews in a timely manner and give useful feedback on what was done well, helpful suggestions for how to improve, and specific comments about why you gave the grade you chose to assist peers in their learning.
- Turn assignments in on time and follow instructions on all assignments including those affecting the use of technology.
- Be truthful in all communication, which includes, but is not limited to, avoiding academic dishonesty.

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